

Self-supervised Vision Transformers for Geometric 3D Reconstruction and Biomass Estimation

Abstract—Above-ground biomass (AGB) estimation is crucial for carbon accounting and climate change monitoring, as well as for agricultural productivity and plant health. Non-invasive methods leveraging LiDAR sensing and deep learning have emerged to predict AGB without destructive sampling. Recent models demonstrate progress using 3D point clouds requirements, rigid model architectures, and poor transferability across heterogeneous vegetation and sensing conditions. The approach presented in this study integrates volumetric estimation using a vision transformer operating on multi-resolution voxel grids and visual geometry grounding that establishes explicit correspondences between image features and three-dimensional coordinates through cross-attention mechanisms. Notably, in regions with complex branching structure, our method shall reduce geometric distortion as measured by surface consistency. While this approach shall excel in capturing branching structures and foliage distribution, limitations remain in reconstructing very fine features and highly reflective surfaces. The resulting 3D models facilitate precise quantification of architectural traits enabling biomass estimation and robotic teleoperations on plants.

Keywords—biomass estimation, 3D image reconstruction, visual geometry grounding, self-supervised vision transformers.

I. INTRODUCTION

Automated estimation of plant biomass is essential for environmental monitoring, carbon accounting, and sustainable land-use management [1], [2], [3], [4], [5], [6], [7], [8], [9]. Some plants such as trees are natural carbon sinks and fundamental in regulating the Earth's climate. Biomass estimation enables us to accurately quantify carbon storage, model tree growth and implement sustainable use case strategies for land. For crops, biomass enables us to understand plant physiological conditions and monitor the growth cycles for effective planning. Remote sensing technologies, particularly Light Detection and Ranging (LiDAR), have emerged as powerful tools for capturing detailed structured information from vegetation, enabling non-invasive, scalable biomass modelling [5], [10], [11], [12], [13], [14]. However, current LiDAR-based approaches often suffer from limitations in generalizability, and efficiency especially when deployed to heterogeneous plant structures[9].

This research addresses the critical gap between 2D image acquisition and accurate 3D geometry reconstruction of plants to estimate biomass by proposing a novel framework that integrates volumetric transformers with visual geometry grounding [15]. The key challenges we tackle include; (1) developing representations that effectively capture the multi-scale nature of plant structures from coarse overall shape to fine branching patterns; (2) designing attention mechanisms that can leverage geometric constraints to improve reconstruction fidelity; (3) creating a learning framework that generalizes across plant species and growth stages despite limited training datasets and data labels; (4) establishing methods for accurate volumetric estimation to correlate with plant biomass. Unlike

previous approaches that treat geometry principles into the transformer architecture, enabling more physically plausible and botanically accurate reconstructions [12], [16], [17], [18]. This integration is essential for agricultural applications where precise geometric understanding directly translates to improved crop management decisions and resources allocation.

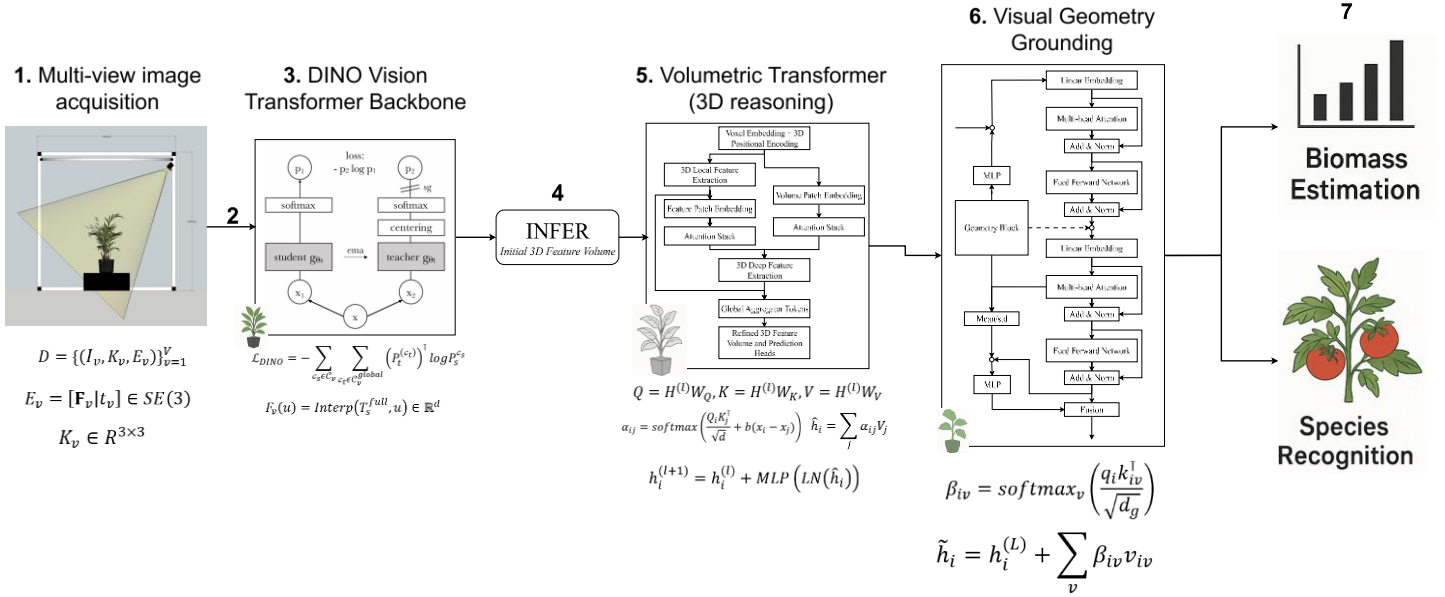
To overcome these limitations, this research presents a dual-transformer architecture for high-fidelity three-dimensional reconstruction of a plant from multi-view two-dimensional images to address agricultural automation where complex occlusions and irregular geometries have impeded progress. The hybrid architecture enables robust geometric reasoning including regions of variable illumination unlike relying solely on implicit neural representations [19], [20]. Evaluations on a diverse dataset comprising multi-view images at various growth stages demonstrate significant improvements over state-of-the-art methods. This approach is to be evaluated across mean Intersection over Union (IoU), Chamfer distance, and F-score, and outperform NeRF-based approaches [19] and point-based methods respectively [21].

The contribution is multifaceted. First, a novel volumetric transformer architecture that efficiently encodes spatial relationships within plant structures while maintaining computation tractability is introduced. The architecture extends previous work on vision transformers [22], [23] by incorporating volumetric attention mechanism specifically designed for plant morphology. Second, we propose integrating a visual geometry grounded technique that leverages botanical priors to constrain the reconstruction process [24], [25], significantly improving accuracy in areas with sparse visual information. This approach builds upon recent advances in physic-informed neural networks [26] but adapts them to the specific context of plant geometry. Third, we develop a hybrid loss function that balances geometric fidelity with botanical plausibility, addressing a key limitation in existing methods that often produce structurally implausible reconstructions [21]. Finally, we contribute a comprehensive dataset of paired 2D images and ground-truth 3D models of various plants at different growth stages, which addresses the scarcity of high-quality training data in this domain [27], [28].

II. METHODOLOGY

The pipeline has four stages: (1) calibrated multi-view image acquisition; (2) a DINO visual transformer backbone that extract high-dimensional features and preserves long-range spatial dependencies; (3) projection of multi-view features into a 3D feature volume using camera parameters and multi-view geometry; and (4) a volumetric transformer that reasons directly in voxel space to infer an occupancy probability for each grid cell, handling occlusions and complex branching. A geometry-aware refinement module then aligns image features with predicted 3D structure via cross-attention and imposes differential-geometry constraints and botanical priors (e.g., branch connectivity and growth patterns). This refinement

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markedly improves thin structures such as stems and fine branches, which are challenging for volumetric methods alone. The system outputs a high resolution, botanically plausible 3D model suitable for downstream agricultural tasks including biomass estimation and robotic teleoperations.

The system architecture is illustrated in the diagram above.

1. A fixed rig with a circular turnable arc is used to collect repeatable baselines and stereo-temporal redundancies. 3 to 5 boards around plant volume are added for scale and pose anchors. Recordings of RGB and aligned Depth with camera intrinsics at 30fps are taken. The frames are saved per intrinsic (f_x, f_y, c_x, c_y) and initial extrinsic via tag pose from the rig are taken too. The depth is preset to high accuracy. For occlusion heavy canopies, capture of 360° seep at multiple elevations. The frames are initialized from the rig kinematics and short checkerboard sequences used to refine camera intrinsics for the camera. Bundle adjustments to jointly optimize $E(R|t)$ and K over the sequence with sparse features for reprojection priors to yield metrics with multi-view consistent poses. The dataset of each frame taken shall this be $C = \{(I_i, K_i, E_i)\}$ which when combined together shall result into $D = \{(I_v, K_v, E_v)\}_{v=1}^V$.
2. During preprocessing, depth cleanup by hole filling, temporal median and bilateral smoothing is performed. The plant mask with learned segmenter is used to enforce multi-view mask consistency via silhouettes. The 2D tokens are patched from masked RGB-D data and 3D tokens initialize multi-resolution voxel grid per voxel feature used for occlusion rendering.
3. Distillation with No Labels (DINO) is applied as the Vision Transformer backbone [25], [29], [30]. DINO supports feature extraction through the use of a teacher-student model to produce tokens $T_v = \Phi_\theta(I_v) \in \mathbb{R}^{N_v \times d}$ from the encoder Φ_θ . These are up-sampled to a dense feature map $F_v: \Omega_v \rightarrow \mathbb{R}^d$ which is bilinear from tokens.

These therefore support self-supervised learning in the pipeline from multi-crop augmentations and output dense pixel/patch features.

4. At the Inference stage, multi-view projection to a 3D feature volume is performed with the canonical voxel grid $\mathcal{G} = \{x_i\}_{i=1}^M$ (voxel centres) aggregating per voxel feature by sampling each F_v at the projected location and weighting by visibility/confidence term $w_v(x_i)$.
5. The Volumetric transformer applies stacked blocks of windowed 3D attention (shifted windows like in Swin Transformer [23]), ray-slice attention along camera rays and cross scale fusion. The initial voxel tokens is expressed as $h_i^{(0)} = W_{in} V_i^{(0)} \in \mathbb{R}^d$ for layer $l = 0 \dots L-1$. The output is the predicted occupancy/density $\sigma_i = \sigma(w_\sigma^T h_i^{(L)})$, $c_i = W_c h_i^{(L)} \in \mathbb{R}^3$. The symbol $\sigma(\cdot)$ is sigmoid where σ_i is interpreted as density at voxel x_i .
6. Geometry-aware refinement from visual geometry grounding applies cross attention alignment with image features for 3D tokens. For each voxel x_i , the per-view features at the voxel's projects are gathered. The geometry-aware constraints from differential geometry and botanical priors will let the continuous occupancy field be computed for gradients and curvature $\nabla \sigma(x) \approx FD(\sigma)$, $\Delta \sigma(x) = \nabla \cdot \nabla \sigma(x)$.
7. The training objective for biomass estimation is derived from the compact plant volumes. The output and evaluation follows mesh extraction $\mathcal{M} = MC(\{\sigma_i^* \geq \tau\})$ for marching cubes at threshold τ . The Intersection of Unions with ground truth voxels, Chamfer-L2 surfaces, F-scores and PSNR/SSIM.

III. CONCLUSION

This work proposes coupling DINO pretraining with volumetric transformers and geometry-grounded refinement yields botanically precise 3D reconstructions from sparse and

views. By unifying multi-view feature lifting, occlusion-aware 3D reasoning, and differential-geometry priors, the pipeline could capture thin branches and complex canopies that defeat NeRF or CNN only baselines. Differentiable rendering will allow supervision learning without 3D ground truth, while priors may enforce connectivity and plausible growth structure. The resulting meshes would then support downstream agronomy, from biomass estimation and automated teleoperations. Overall, the architecture might advance plant-scale reconstruction toward robust, data efficient, and deployable tools for agricultural-based decision-making and automated farm teleoperations.

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